

Failures of Contingent Reasoning in Screening Decisions[‡]

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September 2, 2026

Abstract

Screening decisions—choosing a costly action that reveals information about an unknown quality—can be crucial in various situations, such as hiring workers or designing policies. Choosing whether to screen can be difficult because it requires recognizing and trading off the benefits of information with the costs of getting it. We investigate whether people recognize when the benefits of screening dominate its costs and what causes them to choose otherwise. Our online experiment simulates a hiring scenario with an initial trial task. Participants can first reveal candidates’ quality at a small cost and then choose which candidate to hire. Most participants choose the dominated task that does not reveal the candidates’ quality, and this mistake persists even with experience and feedback. We show that this insufficient screening is driven by failures to anticipate inference and plan the full strategy.

*We thank B. Douglas Bernheim, Lukas Bolte, John Conlon, Stanton Hudja, Muriel Niederle, Kirby Nielsen, Alvin Roth, Jason Somerville, and seminar participants at BABEEW, BEESCuitS, ESA, IZA, and Stanford University for their helpful comments. This study is covered under Stanford’s IRB Protocol 44866 and was registered on the AEA RCT registry under ID AEARCTR-0012230 titled “The inference cost of interventions.”

[†]Data and reproduction materials are available in the online repository.

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I. INTRODUCTION

Screening—choosing a costly action that reveals information about an unknown quality—is important and prevalent in many economic environments. When hiring or promoting employees, selling insurance, or buying a good of unknown quality, individuals and organizations take costly steps to learn more about the other party, like offering them multiple contracts at different prices. In many other situations, however, people do not screen and act with limited information.

The screening decision involves a trade-off: effective screening may be costly, but it reveals valuable information.¹ Assessing whether the informational gains outweigh the costs requires evaluating multiple contingencies, which recent literature shows people have difficulty doing (Niederle and Vespa, 2023). Consider an employer deciding whether to assign a difficult or an easy trial task to a new hire. Difficult tasks, if failed, may come at a greater cost for the employer, but they allow the employer to gauge the new hire’s abilities. Similarly, this trade-off in screening is essential for policy choices. A stringent policy may force everyone to behave the same way, which impedes drawing inferences about them from their behavior. Failures of contingent thinking can, then, make it hard to strike the optimum balance between the forces behind this trade-off, leading to dominated policies.

In his paper, we hypothesize and test that failures of contingent reasoning lead people to screen too little: the screening *costs* are immediate and evident, while the *benefits* of information are only realized in subsequent and contingent decisions, which makes them harder to recognize. We test this hypothesis using an online experiment which mimics a hiring problem: participants see two computer candidates—one Good, one Bad—and must hire one. Before hiring, participants choose a trial task. The Screening task reveals which computer is Good at a small cost; the Pooling task does not. We parametrize the experiment

¹We use “screening” and “pooling” to describe the information content of actions: a screening action induces different-quality agents to produce different outcomes, thereby revealing their quality, while a pooling action does not. This terminology is motivated by the analogy to mechanism design, where a screening mechanism separates types while a pooling mechanism treats all types identically. We do not study screening in the sense of Spence’s signaling framework.

so that screening is dominant for all but extremely risk-loving agents, verified by direct lottery elicitation.²

We run the experiment online and find evidence that people screen insufficiently. In the first round, 68% of participants in the *Baseline* treatment mistakenly choose the Pooling task, compared to only 18% in the *Benchmark* treatment, which provides substantial decision aids and serves as a noise benchmark. Feedback and experience reduce mistakes, but only partially. Two mechanism treatments show that insufficient screening is driven by failures to anticipate inference and to plan the full strategy: each treatment accounts for about half of the non-noise first-round mistakes, and addressing either mechanism is sufficient for participants to learn to screen nearly optimally.³

We provide three main contributions. First, we add to the evidence on failures of contingent thinking (Esponda and Vespa, 2014; Martínez-Marquina et al., 2019; Niederle and Vespa, 2023; Esponda and Vespa, 2024) by showing that they extend from exogenous hypothetical events to contingencies arising from an individual’s own choices. This extension is not trivial, since one might expect self-generated contingencies to be especially difficult to ignore. Our paper thus contributes to the literature showing that individuals make mistakes when they must reason across multiple hypothetical scenarios. While existing theory does not imply that such failures should be limited to exogenous hypothetical events, direct evidence on contingencies generated by one’s own choices remains scarce. One exception is Chakraborty and Kendall (2026), who show that agents fail to perform backward induction when they must respond to their own future decisions. In our setting, however, participants do not mispredict their future actions; instead, they fail to anticipate how their current actions affect the information available to their future selves. Second, we show that having realized rather than hypothetical events does not eliminate the problem: the Good and Bad computers in our experiment are realized, resembling the deterministic treatments that

²Hence, we do not mean the term “dominant” in the statistical stochastic dominance sense.

³The results also hold after accounting for risk preferences, excluding participants whose choices indicate confusion or mistaken information use, and controlling for demographic imbalances across treatments (Section IV).

help resolve failures of contingent thinking in Martínez-Marquina et al. (2019), yet participants make substantial mistakes.⁴ Third, we identify two specific mechanisms—Failure to Anticipate Inference and Failure to Plan—that drive underscreening in the context of a hiring decision. This complements, for example, Araujo et al. (2021), who show that subjects fail to account for how market dynamics endogenously reveal quality (a form of exogenous screening). Our results complement theirs by showing that a similar failure to appreciate the informational value of screening can extend to other settings, like hiring scenarios, where information generation is tied to the agent’s own choice. This connection sets the stage for studies that can test for these mechanisms in applied environments, contributing to a limited literature on cognitive biases in screening (e.g., Bhargava et al., 2017 study health insurance choices, and Tergiman, 2024 study college admissions problems).

Our paper also connects to the experimental literature on learning and bandit problems, which studies the exploration-exploitation trade-off (Bergemann and Välimäki, 2018; Slivkins et al., 2019). Empirically, there is no consensus on whether people under-explore or over-explore. Some papers find evidence of under-exploration (Anderson, 2012; Banovetz, 2020; Hudja and Woods, 2024), while other do not (Kwon, 2020; Hoelzemann and Klein, 2021). We contribute by providing further evidence of under-exploration in a setting where information acquisition is framed as a hiring decision, and by identifying mechanisms that contribute to it.

We also relate to the behavioral literature on failures of strategic reasoning (Milgrom and Roberts, 1986; Eyster and Rabin, 2005; Esponda and Pouzo, 2017; Dal Bó et al., 2018; Eyster, 2019; Calford and Cason, 2024). We contribute by showing that people fail to consider inference and plan their strategy even in a *non-strategic* setup. In particular, experimental research has shown that people fail to use backward induction in strategic games (Johnson et al., 2002; Binmore et al., 2002; Levitt et al., 2011; Dufwenberg and Van Es-

⁴While the existence of a Good and a Bad computer is deterministic, their identity in part 2 is not. It is possible that this is the contingency that participants struggle with, emphasizing the extent to which failures of contingent thinking can manifest.

sen, 2018). One of the mechanisms we identify in our experiment—Failure to Plan—is a non-strategic analog of failure of backward induction, demonstrating that the difficulty of backward induction extends beyond strategic settings.

The rest of the paper is structured as follows. Section II describes the main experimental design and how we test for the mechanisms driving behavior. Section III summarizes the main results and the evidence for the mechanisms. Section IV shows the robustness of the results. Section V discusses some implications of the results and concludes.

II. EXPERIMENTAL DESIGN

We design an experiment to test whether people screen enough. Our online experiment mimics a hiring decision with an initial trial task stage, in which participants choose between two options, only one of which reveals valuable information at an implicit cost. We use a between-subject design with four treatments: *Baseline*, *Benchmark*, *Automatic Inference*, and *Plan*. The first two treatments identify the prevalence of screening mistakes (sections II.D); the latter two test for the mechanisms driving them (section II.E). In the following subsections, we first describe the common experimental setup and then detail each treatment.

II.A. Setting

The experiment needs to have two crucial features to test our hypothesis. First, information must be *instrumental*—that is, it must have decision-relevant value, enabling participants to make better choices. Second, there must be a *screening* decision: the participants need to choose a policy where one option provides valuable information—at some cost—while the other does not. This creates the key trade-off in our hypothesis. Do participants mistakenly choose not to screen?

Our experimental design incorporates the two features that we describe above. We frame

the experiment as a hiring problem because it is a naturalistic setting where inference is essential. The participant faces two computers and needs to decide which one to hire. One of the computers is of *Good* quality, and the other is of *Bad* quality. Hiring the Good quality computer is dominant, but the participant does not know each computer’s quality. However, the participant can learn the quality by observing the computer’s performance in a trial task. The participant makes a policy choice: they choose whether the trial task is *Pooling* or *Screening*. If the task is Screening, the participant can infer the computers’ quality at an implicit cost. The inference happens by observing the payoff that the computer produces when solving the task. If the task is Pooling, they cannot infer anything.

We use computer players rather than human participants as the experiment’s candidates because this rules out confounders and improves identification. First, if candidates were human participants, working on a task would provide experience with that task. This scope for experience could incentivize participants to choose one trial task over the other, which would confound our results. Second, social preferences could affect the participants’ choice of trial task and hiring decision, and the direction of this confounder is ambiguous. Third, we need to ensure consistent performance of candidates across tasks, which is what defines the Good and the Bad candidates. Human candidates’ performance is likely to be inconsistent, which would cause the trial task to reveal little information. We avoid all these concerns by replacing human candidates with computers. There are no training, social preferences, or inconsistency concerns with computer candidates.

We now detail the experiment setup that is common to all treatments.

II.B. Experiment Setup

Each round of the hiring problem consists of two parts: part 1 is the trial task, which is the main decision of interest, and part 2 is the hiring decision. Figure I illustrates the structure of each round. The first part, the trial task, allows the participant to infer which candidate is better. The second part, the hiring decision, makes this information valuable.

Specifically, two computers, Good and Bad, generate a payoff for the participant by solving tasks. There are two types of tasks: Pooling and Screening. In the Pooling task, both computers generate the same high payoff, P_H . In the Screening task, the Good computer generates a high payoff, P_H , while the Bad computer generates a low payoff, $P_L < P_H$. Thus, the participants can infer the quality of each computer if they observe the payoffs they generate in the Screening task.⁵ Choosing the Screening task provides information, but it comes at a cost. Thus, the participant faces a trade-off between receiving a lower payoff from the trial task ($P_H + P_L < 2 \times P_H$) and learning the computers' quality. This trade-off is the crucial element of our design.

After choosing the Pooling or the Screening task in part 1, the participant enters the part 2 hiring decision, in which knowing the computers' quality is valuable because the computers are always hired to solve the Screening task. The stakes in the part 2 Screening task are much higher than in the trial task, which justifies forgoing a part of the payoff in the trial task to learn which computer is Good.⁶ More precisely, when screening is dominant, in part 2 the good computer produces P_G , and the bad computer produces $P_B < P_G$ with $P_G > 2 \times P_H - (P_H + P_L)$ (i.e., the payoff gain from identifying and hiring the Good computer in part 2 is large enough to outweigh the payoff sacrificed by choosing the Screening task rather than the Pooling task in part 1). Hence, if the participant chooses the Screening task in part 1 and hires the Good computer in part 2, they get $P_H + P_L$ in part 1, and P_G in part 2, for a total payoff of $P_H + P_L + P_G$ with certainty. If the participant instead chooses the Pooling task in part 1, they receive $2 \times P_H$ in part 1 and face an equally likely lottery over P_G and P_B in part 2, for total payoffs of $2P_H + P_G$ or $2P_H + P_B$, each with probability 1/2. We choose parameters such that the certain payoff from Screening exceeds the expected payoff from Pooling—that is, $P_H + P_L + P_G > 2P_H + (P_G + P_B)/2$, which reduces to $P_G - P_B > 2(P_H - P_L)$ —which ensures that Screening is dominant for any risk-neutral or risk-averse participant,

⁵That is, if they observe a payoff of P_L (Screening task), they know that the computer that generated it is Bad. If they observe both computers generating P_H (Pooling task), then they cannot infer the computers' quality.

⁶We discuss mistakes in the trial task in subsection II.C.

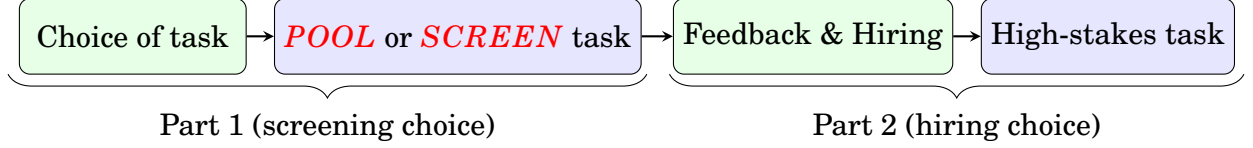


Figure I: Diagram of experiment for Screening-dominant rounds

and even for moderately risk-loving ones.⁷

Participants face ten rounds with varying payoff parameters. Table I summarizes the parameter values in the ten parameterizations. The parameterization in the first and last rounds are the same for all participants, but parameterizations appear in random order in rounds 2–9. Parameterizations 1–6, which include the first and last rounds, follow the description above, where it is dominant to choose the Screening task.⁸ Parameterizations 7–10 make it dominant to choose the Pooling task. While our primary interest is the Screening-dominant rounds, we include Pooling-dominant rounds to prevent participants from mechanically choosing the Screening task without regard for the payoffs. If the risk elicitation (detailed in section II.C) is not randomly picked to determine the participant’s bonus, one randomly picked parameterization counts toward the participant’s payoffs (see section II.F for more details about participants’ payoffs). We label the Pooling and the Screening tasks with color names (e.g., Brown and Blue tasks), and vary them every round.⁹

At the beginning of the experiment, we thoroughly explain its structure to the participants so they have all the information to make dominant choices. We carefully explain what decisions the participants will face, what payoffs they can receive, and what information they will have. Importantly, the participants know that they will observe the performance of each computer before the part 2 hiring stage. We employ rigorous comprehension checks to verify that the participants understand the setup. Figure II shows the main description

⁷The only change needed for the Pooling task to be dominant is that parameters are such that the condition reverses—that is, $P_H + P_L + P_G < 2P_H + (P_G + P_B)/2$, which reduces to $P_G - P_B < 2(P_H - P_L)$.

⁸For example, we use parameterization 1 in the first round: In the Pooling task, both computers generate \$0.05. In the Screening task, the Good computer generates \$0.05, while the Bad computer generates \$0.00. In part 2, the Good computer generates \$4.30, while the Bad computer generates only \$0.05.

⁹This further randomizes the positioning of the tasks on the screen, preventing participants from anchoring to an initial option.

Param. #	Part 1			Part 2		
	<i>POOL</i> , both	<i>SCREEN</i> , Good	<i>SCREEN</i> , Bad	Good	Bad	Optimum
1	0.05	0.05	0.00	4.30	0.05	<i>SCREEN</i>
2	0.05	0.05	0.00	4.45	0.10	<i>SCREEN</i>
3	0.20	0.20	0.15	4.30	0.10	<i>SCREEN</i>
4	0.05	0.05	0.00	4.45	0.10	<i>SCREEN</i>
5	0.05	0.05	0.00	4.35	0.10	<i>SCREEN</i>
6	0.05	0.05	0.00	4.50	0.10	<i>SCREEN</i>
7	2.20	2.20	0.15	0.20	0.20	<i>POOL</i>
8	2.10	2.10	0.00	0.05	0.05	<i>POOL</i>
9	2.00	2.00	0.00	0.05	0.05	<i>POOL</i>
10	2.15	2.15	0.00	0.05	0.05	<i>POOL</i>

Table I: Summary of payoff parameters for the ten parameterizations. All values are in USD terms. The parameterization in the first and last rounds are the same for all participants, and randomly drawn from parameterizations 1–6. In rounds 2–9, parameterizations are randomly drawn from all remaining ones.

of the two parts in the *Baseline* treatment. All experimental instructions are in Online Appendix section A.B.

II.C. What is a mistake?

Parameterizations 1–6: Participants’ risk preferences determine whether Pooling is a mistake or not. The two part 1 trial tasks induce two different lotteries, and we show that participants’ risk preferences are such that choosing the Pooling task in the Screening-dominant rounds is indeed a mistake.¹⁰

To illustrate the payoff consequences of a Screening-dominant round, consider the first round parameterization (row 1 of Table I), which is the same for everyone: If the participant chooses the Screening task in part 1 and hires the Good computer in part 2, she gets \$4.35 with certainty (\$0.05 + \$0.00 in part 1 and \$4.3 in part 2). If the participant instead chooses the Pooling task in part 1, then in part 2 she faces a lottery with equally likely payoffs

¹⁰While participants make choices sequentially, time preferences over monetary payoffs do not affect the choice of the trial task because all payments happen at the end of the experiment. Moreover, this is a single-person decision problem, so strategic considerations play no role.

Decision Green/Yellow: Computers solve Green or Yellow Tasks

You will go through ten decisions. Instructions for all decisions are similar, but the underlined numbers on this page change between decisions.

There are **two Computers** which solve tasks. There are two tasks: **Green task** and **Yellow task**. One computer is of **Good quality**, and the other is of **Bad quality**.

Part 1

The computers solve a task. **You decide which task the computers solve.**

Each computer produces money from solving the task. How much money it produces depends on its quality as follows:

-  **Good quality:** Produces \$0.05 in the Green task and \$0.05 in the Yellow task.
-  **Bad quality:** Produces \$0.05 in the Green task and \$0.00 in the Yellow task.

Bonus from Part 1: You get the amount of money the computers produce in the task.

Before part 2, you will see how much money the computers produce. This is the only extra information you will see.

Part 2

The same computers solve the Yellow task.

Each computer produces money from solving the task. How much money it produces depends on its quality as follows:

-  **Good quality:** Produces \$4.30 in the Yellow task.
-  **Bad quality:** Produces \$0.05 in the Yellow task.

Bonus from Part 2: You will *choose* which computer determines your bonus. You get the amount of money the computer of your choice produces in the task.

Next

Figure II: *Baseline* treatment main instructions.

of \$4.40 (\$0.05 + \$0.05 in part 1 and \$4.3 in part 2) and \$0.15 (\$0.05 + \$0.05 in part 1 and \$0.05 in part 2).¹¹ Hence, the Screening task is dominant unless the participant is

¹¹In Screening-dominant rounds, the Pooling task reduces the participants' expected payoff by at least \$2 compared to the Screening task, which represents 53% of their average bonus payment.

sufficiently risk-loving.¹²

We diagnose the strength of our assumption that the Screening task is dominant by directly eliciting participants’ risk preferences. Assuming that a direct choice over lotteries better captures participants’ risk preferences, this exercise supports our claim that choosing the Pooling task is indeed a mistake.¹³ We elicit participants’ preferences over the lotteries induced in each parameterization by, after the main experiment, having them face six binary menus of lotteries. Each menu has two lotteries induced by one of the six Screening-dominant parameterizations: one lottery is a fixed payoff resulting from choosing the Screening task in part 1 and hiring the Good computer in part 2, and the other is a lottery resulting from choosing the Pooling task in part 1 and hiring a computer randomly in part 2. Figure A.1 in the Online Appendix summarizes the results. Across all six parameterizations, less than 4% of participants choose the lottery induced by the Pooling task.

Parameterizations 7–10: The Pooling task produces a certain payoff that is strictly larger than the certain payoff produced by the Screening task, unambiguously constituting a mistake.

II.D. Baseline and Benchmark Treatments

We use two treatments to show that participants make screening mistakes: the *Benchmark* treatment, which estimates the amount of noise in the participants’ answers, and the *Baseline* treatment, which identifies insufficient screening beyond that which comes from our noise benchmark.¹⁴

¹²Assuming CRRA utility, if a person prefers the lottery induced by the Pooling task, they would also prefer a lottery that pays \$101.15 and \$0 with equal probability over a certain payment of \$100.

¹³While we find this to be a reasonable assumption in this case, and approaches like this one are not rare in experimental economics, we recognize the exercise suffers from a “circularity trap,” by which we identify bias by looking for choices that conflict with true preferences while inferring true preferences from unbiased choices (Bernheim and Taubinsky, 2018).

¹⁴For the *Benchmark* to be a good benchmark, we assume that the noise in responses to the *Baseline* and *Benchmark* treatments is similar.

Baseline treatment: We design this treatment to mimic the natural order of choices in a naturalistic hiring scenario: The participants first choose the trial task, then move on to the hiring stage, where they observe the payoff each computer generates and choose which one to hire. As in any experiment, we expect some participants to make the mistake of choosing the Pooling task for irrelevant reasons, such as lack of attention and trembling hand errors, which we interpret as experimental noise that artificially increases the rate of mistakes. Thus, the mistake rate we observe in the *Baseline* treatment combines the screening mistakes caused by stable cognitive errors—which we are interested in—with mistakes stemming from noise. We separate the two to measure the true prevalence of screening mistakes using the *Benchmark* treatment.

Benchmark treatment: We design this treatment to measure the share of mistakes that come from experimental noise. This control treatment is meant to help participants as much as possible while keeping the structure of the experiment the same. We use the share of participants who still make dominated choices in such a scenario as a measure of noise. In this treatment, participants submit their whole strategy, which requires solving the problem backward. They start with the part 2 hiring decision and choose which computer to hire in each contingency, when the part 1 trial task is the Screening task (i.e., when they know the computers’ quality) and when the part 1 trial task is the Pooling task (i.e., when they do not know the computers’ quality).¹⁵ Next, they complete the part 1 trial task. At this point, we remind them about their contingent decisions for the hiring stage. The participants also see the payoff consequences of their choices next to the two task options, which provides no new information but helps them conveniently assess the two options. Because the *Benchmark* is such a strong intervention, we attribute any mistakes in this treatment to inherent noisiness and use it to establish the benchmark amount of noise in the experiment.¹⁶

¹⁵Specifically, in choosing for the contingency in which the part 1 trial task is the Screening task, the participants see the labels for the Good and Bad computers before making their hiring decision.

¹⁶Note that to the extent that this treatment is not strong enough to remove the cognitive bias that drives mistakes, the *Benchmark* will overestimate noise. We can then interpret our estimates as lower bounds for the prevalence of the screening mistakes we study.

Treatment	Key Difference from <i>Baseline</i>	Purpose
<i>Baseline</i>	—	Measure total mistake rate
<i>Benchmark</i>	Backward elicitation; payoff info displayed	Noise benchmark
<i>Automatic Inference</i>	Computer quality revealed if Screening chosen	Test: Failure to Anticipate Inference
<i>Plan</i>	Simultaneous choice of full strategy	Test: Failure to Plan

Table II: Summary of treatments. All treatments share the same experiment structure (Section II.B). Between-participants design; each participant is randomly assigned to exactly one treatment.

II.E. Mechanisms: Automatic Inference and Plan Treatments

By our experimental design, the mistake of choosing the Pooling task cannot be attributed to standard explanations of time inconsistency over payoffs, risk aversion or mistakes in strategic reasoning. To solve the problem correctly, participants must *recognize* and *use* the informational content in the computers’ performance on the part 1 Screening task. We separately test for both — whether they fail to anticipatedly recognize the inference they will be able to make (henceforth, “Failure to Anticipate Inference”), and whether they fail to use it in planning their strategy (henceforth, “Failure to Plan”). To test Failure to Anticipate Inference, we design the *Automatic Inference* treatment, in which we tell the participants that we will make the inference for them. To test Failure to Plan, we design the *Plan* treatment, in which we make participants choose their full strategy from the beginning. These interventions substantially reduce the mistakes in Screening-dominant rounds, which supports Failure to Anticipate Inference and Failure to Plan as relevant mechanisms driving underscreening.

The Failure to Anticipate Inference suggests that participants do not know that they will be able to make inferences in the future: they do not realize that the Screening task provides information that reveals each computer’s quality. Despite this, they may be thinking about part 2 and planning their strategy for the future. The *Automatic Inference* treatment, a

light-touch intervention to the *Baseline* treatment, tests for this mechanism.

Automatic Inference treatment: This treatment is different from the *Baseline* treatment in one crucial way: instead of participants making the inference themselves, we make the inference automatically for them. If the participant chooses the Screening task, we explicitly tell them the quality of each computer before their hiring decision. If the participant chooses the Pooling task, we do not reveal the computer quality. Participants know this when choosing between the Screening and Pooling tasks in part 1. Specifically, next to the Screening task option in the experiment interface, we explain that we will reveal the computers’ quality; next to the Pooling task option, we explain that we will not. If the hypothesized Failure to Anticipate Inference mechanism is correct, this additional message at the first stage should significantly reduce the mistake rate.

By our second hypothesized mechanism, Failure to Plan, participants do not plan their entire strategy when choosing the trial task; instead, they myopically focus on the immediate choice in front of them. We design the *Plan* treatment to test for Failure to Plan.

Plan treatment: This treatment changes how we elicit participants’ strategy relative to the *Baseline* treatment. We ask participants to choose a complete plan for their strategy from the beginning. Specifically, they choose between three options: (i) choosing the Screening task and hiring the computer that produces the larger amount, (ii) choosing the Screening task and hiring the computer that produces the smaller amount, and (iii) choosing the Pooling task and hiring one of the computers chosen randomly. By simultaneously choosing for parts 1 and 2, this elicitation forces participants to consider the whole problem. If the Failure to Plan mechanism is causing mistakes, this planning tool should reduce the rate of mistakes relative to the *Baseline* treatment.

II.F. Procedures: Online Experiments on Prolific

We recruited all participants on Prolific, an online platform frequently used for research studies, on October 10th 2023 and April 25th 2024. We restrict the sample to participants in

the USA who are fluent in English and have completed at least 100 previous submissions on Prolific, with a minimum approval rate of 97%. The experiment was implemented using the oTree platform (Chen et al., 2016). The study was registered on the AEA RCT registry with ID AEARCTR-0012230 under the title “The Inference Cost of Interventions.”

We recruited 982 participants who were randomly assigned to the four treatments: 251 subjects were assigned to the *Baseline* treatment, 236 to *Benchmark*, 244 to *Automatic Inference*, and 251 to *Plan*.¹⁷ In all treatments, for each participant, one of the decisions (including the lottery choice decision in the risk elicitation) is randomly selected and the outcome of that decision is used to determine the participant’s bonus. Participants receive a \$3 completion payment and an average bonus payment of \$3.79, and the median completion time is 21 minutes, which is equivalent to \$19.39 per hour.

We follow the standard procedures to ensure that the results are not driven by misunderstanding of the experiment instructions. Participants receive detailed instructions and must correctly answer a set of understanding questions about them before proceeding, as well as passing attention checks. Figure A.2 in the Online Appendix shows the distribution of mistakes on the understanding questions.¹⁸ Restricting the sample to participants who make zero mistakes does not affect the results.¹⁹ Importantly, we present participants with parts 1 and 2 on the same page, so that it is unlikely that the failure to consider part 2 in their part 1 decision is an artifact of the experiment (see Online Appendix section A.B for screenshots of the experimental instructions).

¹⁷The sample is balanced on gender. The average age is 42 years. 74% of the sample identify as White and 11% as Black. 53% of the sample have a Bachelor’s degree or higher. Table A.1 in the Online Appendix summarizes demographic characteristics by treatment.

¹⁸For example, participants cannot proceed in the experiment without correctly answering that the computers in part 1 and part 2 are the same.

¹⁹Additionally, after the *Baseline* participants make their part 1 decision, we ask them whether they will see how much money each computer produced in part 1. Around 71% of the participants answered this question correctly on the first try, confirming that most participants understand this essential aspect of the experimental design (restricting to only these participants does not affect the results, as we discuss in section IV).

III. RESULTS: PREVALENCE OF MISTAKES

The *Baseline* and *Benchmark* treatments show that people fail to account for the inference consequences of their policy choices. Most participants—68%—mistakenly choose the Pooling task, and the mistakes do not disappear entirely with experience. The *Automatic Inference* and *Plan* treatments substantially reduce the mistake rate, suggesting that the Failure to Anticipate Inference and Failure to Plan mechanisms are important drivers of insufficient screening. Participants do not recognize they can extract valuable information in the future, and do not plan the entire strategy.²⁰

III.A. *Baseline and Benchmark Treatments*

Choosing the Pooling task is a mistake in six of the ten parameterizations. Since our hypotheses concern the Screening-dominant rounds, where failures of contingent thinking can lead to underscreening, we focus on these six parameterizations in the rest of the analysis.²¹ We expect participants to learn over time from feedback and experience. Therefore, we present the main results separately by round.

Figure III shows the main results. In the first round, most participants screen insufficiently: 68% of participants choose the Pooling task in the *Baseline* treatment. In the *Benchmark* treatment, only 18% choose the Pooling task. The difference is statistically significant (t-test yields a p-value < 0.001). Overall, in the six Screening-dominant rounds, participants make an average of 2.37 and 1.22 mistakes in the *Baseline* and the *Benchmark* treatments, respectively (p-value < 0.001).

Learning: There is substantial learning from experience, but mistakes disappear only partially. In rounds 2–10, the rate of mistakes reduces to an average of 34% in *Baseline*. It

²⁰While we design each treatment to tackle one mechanism, we cannot rule out the possibility that either treatment allows participants to correct both mechanisms (for example, because the interaction of both mechanisms made it too complex for them to realize, but the elimination of one allows for a full correction).

²¹We include Pooling-dominant rounds to prevent participants from mechanically choosing the Screening task without regard for the payoffs. Hence, they serve an auxiliary role since there is no relevant trade-off in these rounds.

is still significantly higher than the rate of mistakes in the *Benchmark* treatment, which is, on average, 21% (t-test yields a p-value < 0.001).²² Most of the learning happens after round 1, with no learning in the *Benchmark* treatment and little learning across rounds 2–10 in the *Baseline* treatment. The difference in the mistake rate between the *Baseline* and the *Benchmark* treatments is statistically significant in nine out of ten rounds. Online Appendix Table A.2 further shows that mistakes are robust to learning by showing that the presence of interspersed Pooling-dominant rounds does not disrupt the learning process; Facing Screening-dominant rounds in a row does not substantially change the learning pattern, with the rate of mistakes in *Baseline* remaining steady at around 40%. Moreover, Online Appendix Table A.3 confirms most learning happens between the first and second rounds, with minimal learning as participants face their third, fourth, fifth, or sixth Screening-dominant round.²³

III.B. Mechanism: Automatic Inference and Plan Treatments

The mistake rate in the *Automatic Inference* treatment is significantly lower than in the *Baseline* treatment. Figure III compares the mistake rates in all treatments. In the first round, only 44% of the participants make the mistake of choosing the Pooling task. This mistake rate is about halfway between the *Baseline* (68%) and the *Benchmark* (18%) treatments. In subsequent rounds, we observe complete learning to the same level as in the *Benchmark* treatment. The difference in the mistake rate from the *Baseline* is statistically significant in round 1 and in rounds 2–10 combined (t-tests yield p-values < 0.001). Considering each round separately, the mistake rate is significantly different from the *Baseline* treatment in seven out of ten rounds. Thus, the *Automatic Inference* treatment suggests that the hypothesized Failure to Anticipate Inference mechanism is important: Participants

²²That mistakes in *Baseline* do not fully disappear with experience is consistent with the experimental evidence in Chakraborty and Kendall, 2026 and that reviewed in Niederle and Vespa, 2023 for failures of contingent thinking.

²³While the rate of mistakes goes down in the last round for *Baseline*, the same happens in the *Benchmark* treatment.

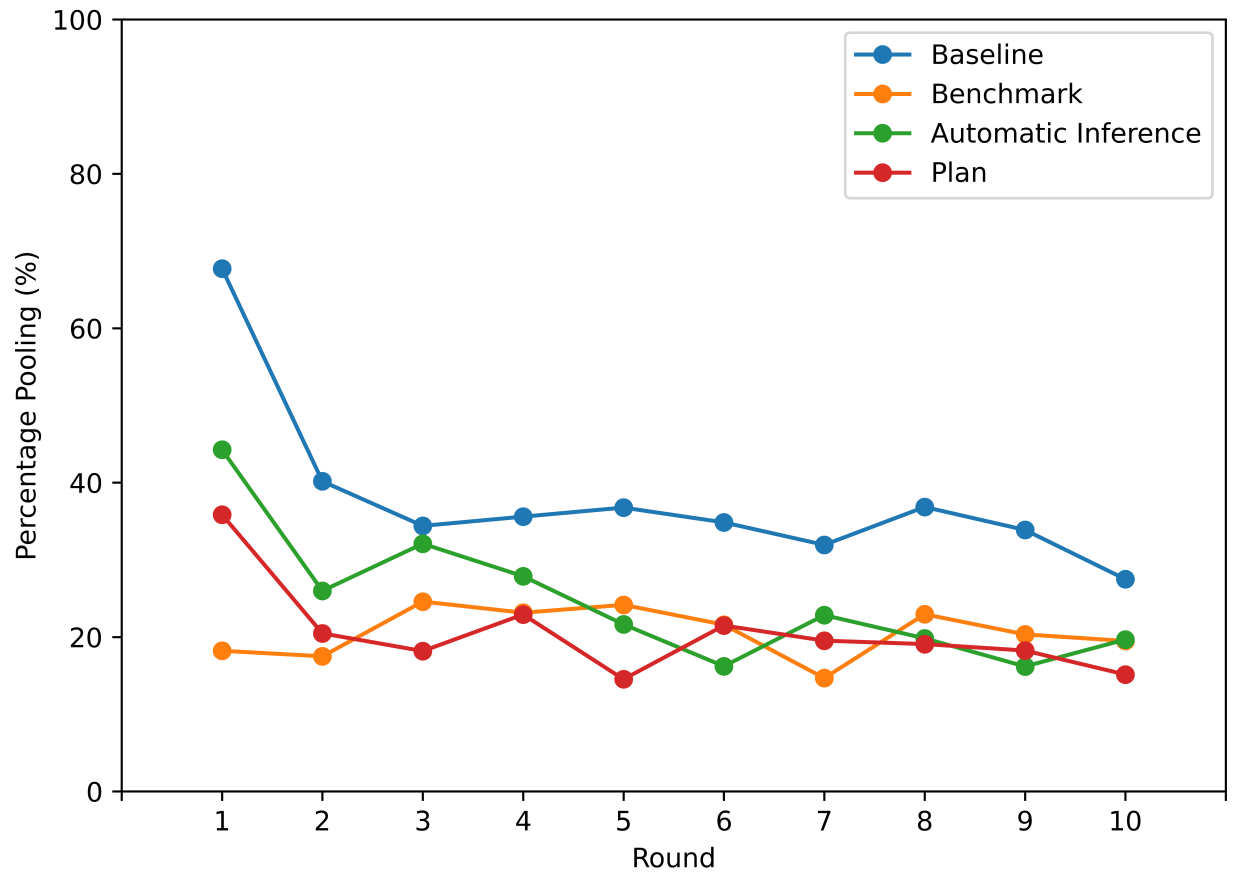


Figure III: Rate of mistakes across rounds with Screening-dominant parameterizations for all treatments.

seem not to understand how to infer information in the future.

Similarly, the mistake rate in the *Plan* treatment is significantly lower than in the *Baseline* treatment (t-tests yield p-values < 0.001). In the first round, 36% of the participants choose the Pooling task.²⁴ In subsequent rounds, participants learn completely — the mistake rate in rounds 2–10 is similar to the *Benchmark* treatment.²⁵

The difference in the mistake rate from the *Baseline* is significant in all rounds. Thus, the *Plan* treatment suggests that the hypothesized Failure to Plan mechanism is important too: Participants do not seem to plan their entire strategy.²⁶ This result is in line with evidence by Chakraborty and Kendall, 2026 supporting the idea that participants fail to think through their own actions in the future, also in the context of an individual decision problem.

All six Screening-dominant parameterizations considered, participants make an average of 1.57 and 1.29 mistakes in the *Automatic Inference* and *Plan* treatments, respectively, compared to 2.37 and 1.22 in the *Baseline* and the *Benchmark* treatments.²⁷ The difference in the average number of mistakes between treatments hides heterogeneity in the distribution of mistakes. For each participant, we calculate the number of mistakes they make in the Screening-dominant rounds (i.e., choosing the Pooling task). Aggregating over all participants within a given treatment, we plot the histograms of the total number of mistakes in Figure IV. In contrast to the other three treatments, a disproportionately large group of participants in the *Baseline* treatment always make the mistake of choosing the Pooling task, and a disproportionately small group of participants never make this mistake.

Participants in the *Baseline* treatment who always choose the Pooling task drive the per-

²⁴The difference between *Automatic Inference* and *Plan* treatments is not statistically significant at the 5% level.

²⁵Online Appendix Table A.3 confirms this learning pattern when focusing on the number of screening-dominant rounds the participants have faced.

²⁶Online Appendix Table A.2 suggests the *Automatic Inference* treatment becomes less effective at reducing mistakes as participants see more Screening-dominant rounds in a row, yet this exercise is much less well-powered, making it hard to draw reliable inferences.

²⁷Relative to the *Benchmark*, the difference is statistically significant for the *Automatic Inference* treatment (p-value < 0.001), but not for the *Plan* treatment (p-value = 0.5).

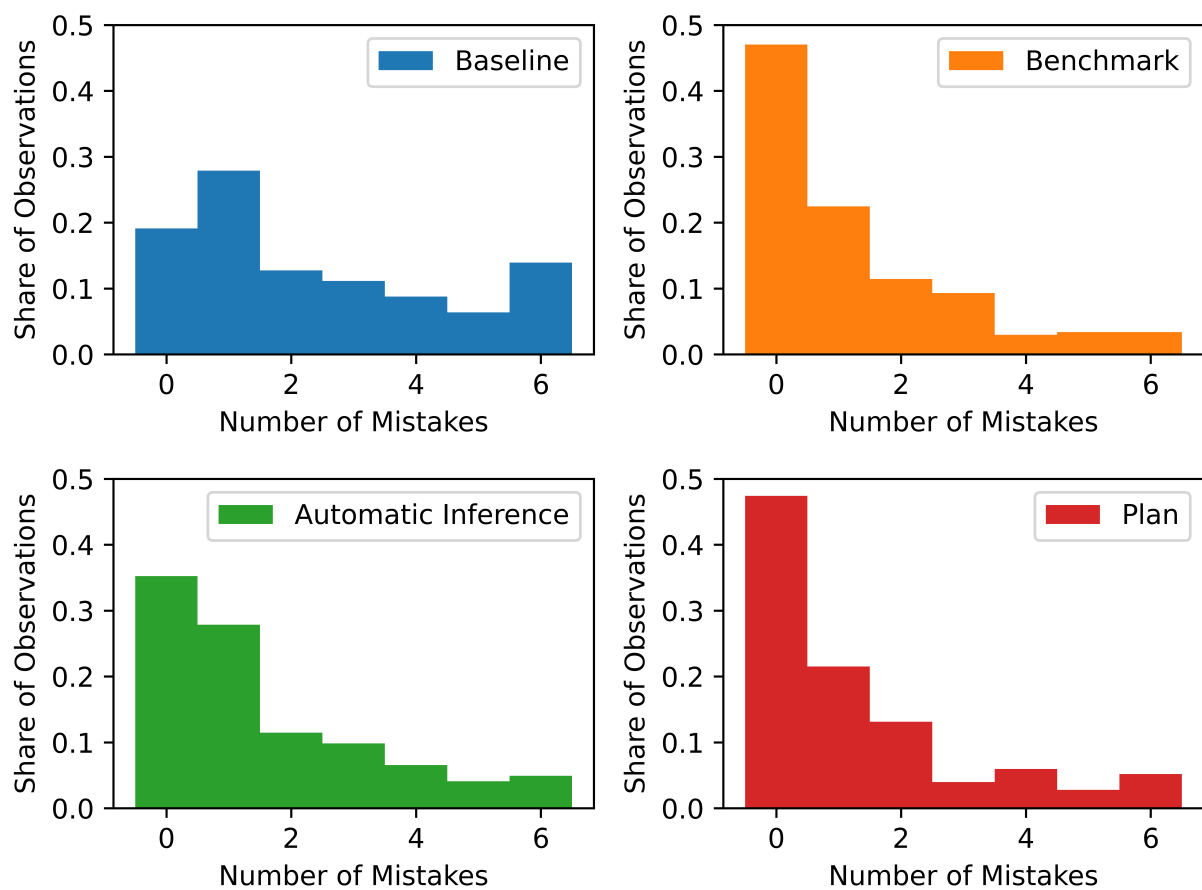


Figure IV: The distribution of total screening mistakes, by treatment

sistence of insufficient screening. These participants never learn that screening is dominant and consistently avoid screening throughout the experiment. If we remove these participants from the sample, round 1 results remain similar, but the mistake rate in rounds 2–10 becomes close to statistically indistinguishable in all treatments except *Plan* (see column 7 in Table III). This result shows that some participants struggle to learn the dominant screening choice without aid, making the mistake rate in the *Baseline* treatment more persistent.²⁸

This section’s results suggest that participants fail to anticipate inference and plan their strategies. These two mechanisms seem to be similarly important drivers of insufficient screening.

IV. ROBUSTNESS OF RESULTS

Table III summarizes the multiple tests we conduct to verify the robustness of our results.

Pooling is a mistake. We directly elicit preferences over the induced lotteries, and almost all participants choose the lottery corresponding to the Screening task. Nevertheless, a few participants sometimes choose the lottery corresponding to the Pooling task, which could explain our results. To rule out this possibility, we construct a subsample without them and re-estimate the magnitude of mistakes. Column (2) of Table III shows that excluding participants who prefer at least one lottery induced by the Pooling task does not affect our results.

Participants’ understanding. Throughout the experiment, participants had to correctly answer a set of understanding questions. If they made an error on an understanding question, they had to correct it before proceeding. This design helps ensure that participants understand the critical features of the experiment even if they missed them while reading instructions.

²⁸Participants who always choose the Pooling task never observe the informational gains from screening. Footnote 31 discusses an alternative treatment that can test for the effect of forced exposure to the gains from screening.

Online Appendix Figure A.2 plots the distribution of errors in understanding questions. Most participants make few errors, with 89% of them making at most one error, and 56% making zero errors. The small number of understanding errors highlights that confusion or misunderstanding was not an issue for most participants. Since error rates are similar across treatments, misunderstanding is unlikely to drive the treatment effects.²⁹ The results are robust to misunderstanding by focusing on the 546 participants who make zero errors on the understanding questions, as Column (3) of Table III shows.

Do participants in the *Baseline* treatment understand that they will observe how much each computer produces before they make the hiring choice? If they fail to understand this, the Screening task is not necessarily dominant. However, asking an understanding question about this at the beginning of the experiment would highlight the connection between the two parts of the problem, potentially shutting down the mechanisms that we are interested in studying. To avoid interfering with the mechanisms, we first elicit participants' round 1 trial task choice and only then ask the understanding question about whether they will observe how much each computer produces.³⁰ Restricting to the 71% of participants who answer this understanding question correctly—they know that they will observe how much each computer produces before the hiring choice—does not affect the results, as we show in column (4) of Table III.

Mistaken use of information. Screening is dominant only if the participants correctly use the revealed information. To claim that choosing the Pooling task is a mistake, we need to show that participants who do so would be able to use the information if they got it. An ideal test for this would be observing participants' counterfactual part 2 hiring choices after the part 1 Screening task. We approximate this counterfactual scenario by looking at participants' actual part 2 hiring choices after choosing the Screening task in part 1. All

²⁹The average number of errors in the *Baseline* treatment is significantly smaller than in the *Plan* treatment and not significantly different from the *Benchmark* and *Automatic Inference* treatments. The distributions of errors are similar across treatments (Online Appendix Figure A.3).

³⁰As a result, the understanding question does not affect round 1, but it may speed up learning, reducing the mistake rate in rounds 2–10 in *Baseline*, which goes against our hypothesis.

Table III: Summary of estimates

Panel A: Round 1 estimates							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Benchmark</i>	-0.50*** (0.04)	-0.51*** (0.04)	-0.53*** (0.05)	-0.49*** (0.04)	-0.49*** (0.04)	-0.49*** (0.05)	-0.47*** (0.04)
<i>Automatic Inference</i>	-0.23*** (0.04)	-0.25*** (0.04)	-0.27*** (0.06)	-0.23*** (0.05)	-0.23*** (0.04)	-0.27*** (0.05)	-0.21*** (0.05)
<i>Plan</i>	-0.32*** (0.04)	-0.34*** (0.04)	-0.39*** (0.05)	-0.31*** (0.05)	-0.32*** (0.04)	-0.30*** (0.05)	-0.30*** (0.04)
<i>Baseline</i> mistake rate	0.68	0.68	0.66	0.67	0.68	0.63	0.62
N	982	934	546	909	982	742	914

Panel B: Rounds 2–10 estimates							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Benchmark</i>	-0.13*** (0.03)	-0.14*** (0.03)	-0.15*** (0.04)	-0.14*** (0.03)	-0.13*** (0.03)	-0.02 (0.03)	-0.05** (0.03)
<i>Automatic Inference</i>	-0.11*** (0.03)	-0.12*** (0.03)	-0.11*** (0.04)	-0.12*** (0.03)	-0.11*** (0.03)	-0.04 (0.03)	-0.05* (0.03)
<i>Plan</i>	-0.15*** (0.03)	-0.17*** (0.03)	-0.17*** (0.04)	-0.16*** (0.03)	-0.15*** (0.03)	-0.07*** (0.03)	-0.09*** (0.02)
<i>Baseline</i> mistake rate	0.34	0.34	0.29	0.35	0.34	0.20	0.23
N	4910	4670	2730	4545	4910	3710	4570

OLS regression results with binary dummies for each treatment (*Baseline* omitted) and mistakes in screening-dominant rounds as dependent variable. Specification by column: (1): Main sample, (2): Participants who never prefer the lottery induced by the Pooling task, (3): Participants who make zero mistakes on the understanding questions, (4): Participants who know that they will observe how much computers produce, (5): Main sample with controls for age, gender, race, and education level, (6): Participants who make zero mistakes in part 2 hiring choice, (7): Participants who choose the Screening task when it is dominant at least once. Robust standard errors in parentheses for round 1 and clustered by participant for rounds 2–10. p -value < 0.1: *, p -value < 0.05: **, p -value < 0.01: ***

Screening-dominant parameterizations are similar, with the only differences being parameters and task names. Therefore, we treat rounds where a participant chooses the Screening task as a counterfactual for the rounds where they choose the Pooling task. Across all treatments, more than 90% of participants hire the Good computer after choosing the Screening task (Online Appendix Figure A.4). This suggests that it is unlikely that participants who do not screen would not be able to use the revealed information.

Although mistakes at the hiring stage differ across treatments, this does not drive the differences in the screening decision. Participants in the *Baseline* treatment make slightly more hiring mistakes than in the other treatments. These differences are too small to explain the large differences in the trial task choices. Nevertheless, we verify that the results are robust to this concern by restricting the sample to those participants who use the information correctly: those who chose the Screening task at least once and always hired the Good computer thereafter. Round 1 results stay the same in this subsample as in the whole sample, although the levels of mistake rates mechanically decrease for all treatments. In subsequent rounds, most treatment effects disappear—participants in the *Baseline* treatment make the same number of mistakes as in the *Benchmark* and *Automatic Inference* treatments, although still more than in the *Plan* treatment (Table III, column (6)). Both the decrease in the levels of mistakes and the faster learning dynamics are expected because this subsample excludes everyone who always chooses the Pooling task, who drive the persistence of insufficient screening in the full *Baseline* sample.³¹

Stakes: The payoff consequences of these mistakes are substantial. By not screening, participants leave on the table at least \$2, which represents 53% of their average bonus payment. While whether these mistakes disappear as stakes increase remains an empirical

³¹An alternative approach to test for the proper use of information among those who do not screen is to exogenously reverse the trial task choice from Pooling to Screening, to randomly force participants who choose Pooling to face the consequences of picking the Screening task. Early pilot results show that most participants hire the Good computer after the trial task choice is reversed from Pooling to Screening. We chose not to implement this approach in the main experiment for two reasons. First, pilot participants report high levels of confusion stemming from this random exogenous reversal, which could drive mistakes. Second, it provides a counterfactual only probabilistically, in a subset of rounds, which would demand a larger sample size.

question (Klein Teeselink et al., 2026), and we do not expect experienced policy-makers or employers to necessarily behave as our Prolific participants, the sample and stakes we study are important in themselves in shedding light on lay people’s behavior in the face of this trade-off, which can determine their support for policies.

First round learning: Many naturally occurring decision problems are effectively one-shot or first-time events. Individuals often lack repeated exposure to such situations, limiting learning opportunities.³² First-round behavior may partly reflect unfamiliarity with the task, including the need to recall which task is which.³³ However, these same conditions characterize many naturally occurring screening decisions. Even if participants later learn, their first-round behavior provides insights into how individuals approach unfamiliar trade-offs in the absence of prior experience.

Nevertheless, it is crucial to understand the persistence of the mistake, and the substantial and immediate drop in the rate of mistakes after the first round calls for caution in interpreting their persistence in our setting. As we argue in our discussion of participants’ understanding, the rate of mistakes does not seem to be driven by confusion or lack of engagement. However, the fact that behavior changes so substantially after round 1, sheds light on the persistence and robustness of the cognitive bias.

We might be estimating an upper bound in learning: As we discuss above, after round 1 we introduce a comprehension question meant to test whether participants in the *Baseline* treatment understand that they will observe how much each computer produces before they make the hiring choice. This question highlights the connection between the two parts of the problem, potentially shutting down the mechanisms that we are interested in studying, which is why we only ask it after round 1. As a result, the understanding question

³²For example, a hiring manager deciding whether to invest additional resources in screening job applicants may face only a few such high-stakes hiring decisions in their career. Similarly, individuals choosing whether to obtain a second medical opinion before a major procedure often do so without prior experience in comparable situations.

³³Later in this section we discuss why we do not believe memory is driving our results. Importantly, the fact that our treatment effects are present in both round 1 and subsequent rounds—where memory of task details could be better—strengthens the case that the mechanisms we identify are genuine cognitive failures rather than mere artifacts of unfamiliarity.

does not affect round 1, but it may speed up learning, leading to an underestimation of the persistence of the mistake.

Moreover, while mistakes exhibit significant persistence, there is heterogeneity in the learning pattern. In contrast to the substantial share of participants who make a mistake only in the first round, Figure IV shows that a disproportionately large group of participants always make the mistake in *Baseline* relative to the other treatments. This heterogeneity suggests that the robustness of the cognitive failure is not globally robust but might rather vary across individuals.

Memory and recall. If participants in *Baseline* forget which task reveals computer quality, the other treatments may reduce mistakes simply by providing reminders rather than by addressing cognitive failures. Several pieces of evidence argue against this interpretation. First, participants undergo detailed instructions and must pass comprehension checks before making their decisions. In particular, in every round, participants must answer “In part 2 of this decision, which task do the computers solve?” before making their part 1 choices (see Online Appendix Figure A.13). Column (3) of Table III shows that the results are robust to focusing on the participants who make zero errors on the understanding questions. Second, in every decision page, participants have access to a “Review Instructions” button which they can click to remind themselves of any information that they might have forgotten or missed (see Online Appendix Figure A.14). Given that the stakes are significant for them, as we mentioned above, we don’t expect participants to continue with the study without knowing crucial aspects of the instructions. Third, the pattern across treatments is inconsistent with a pure memory explanation: if treatment differences were driven by recall, the *Automatic Inference* and *Plan* treatments—which explicitly state the informational content of each task at the point of choice—should eliminate nearly all non-noise mistakes. Instead, it accounts for only about half of the round 1 gap. This pattern is more consistent with genuine cognitive failures than with memory. Hence, we believe the preponderance of evidence supports cognitive failures as the primary driver.

Individual characteristics. We rule out that differences in participants’ self-reported age, gender, race, and education across samples drive the treatment effects. The treatments are balanced on these demographics, but participants in the *Plan* treatment are slightly younger than the *Baseline* participants (Online Appendix Table A.1). Column (5) of Table III shows the results are robust to controlling for these characteristics.³⁴ Moreover, we find no evidence that the heterogeneity in the persistence of the mistake that we note in our discussion of learning after round 1 is related to self-reported age, gender, race, or education.

V. CONCLUSION

We experimentally test, and find evidence consistent with, the hypothesis that failures of contingent thinking lead people to not screen enough when screening is dominant, even without time inconsistency, risk aversion, and strategic considerations. Insufficient screening significantly diminishes, yet persists, with experience and feedback. We further postulate and test two mechanisms that drive insufficient screening. First, participants fail to recognize that screening allows them to make inferences in the future. Second, they fail to plan their entire strategy. We run two additional treatments that test for each of the mechanisms, each of which accounts for about half of the non-noise mistake rate.

Our results identify a specific cognitive bias by which people can systematically underappreciate the informational value of screening. This bias is relevant wherever there is a trade-off between the costs of screening and the benefits of the information it reveals. Consider an illustrative example: if prisons strictly enforce good behavior, every inmate will follow the rules. This policy may reduce violence, but it also prevents learning about inmates’ character and willingness to follow the law, which is information that could be essential for early parole decisions. More broadly, policies that create strong incentives induce uniform behavior, limiting what can be learned from observed behavior. Whether screening

³⁴Controls for age, race, and “other gender,” not shown in column (5) of Table III, are significantly different from zero (e.g., older and black participants are more likely to make mistakes).

dominates pooling in any particular setting is an empirical question that requires careful context-specific analysis beyond the scope of this paper. However, our results suggest that the informational value of screening may be systematically underappreciated, which is a consideration that policymakers and individuals should be aware of when designing or evaluating screening opportunities.

Our findings have important implications that open several avenues for future research.

First, in our experiment, participants make inferences about computers. While this allows us to keep control, an exciting extension would be considering inferences about other people or *the self* in more realistic scenarios. For example, future work can incorporate studies of how failures to appreciate the informational content of educational signals lead to statistical discrimination in hiring (Campos-Mercade and Mengel, 2024). Together with our results, these findings paint a picture of systematic failures in how people handle information in screening and hiring contexts: people fail not only at *using* available information correctly (as Campos-Mercade and Mengel, 2024 show), but also at *acquiring* it in the first place. Moreover, inferences about *the self* play a role in the literature on the unintended consequences of policies (Bitler and Karoly, 2015; Nandi and Laxminarayan, 2016), which shows that external incentives may crowd out intrinsic or ‘warm glow’ motivation (Lepper et al., 1973; Frey and Jegen, 2000; Bénabou and Tirole, 2006; Ariely et al., 2009). Our findings imply that people may not realize that strong incentives limit one’s ability to feel good about themselves based on inferences they can make about themselves, leading to the crowding-out effect.

Our results also have implications for the literature that shows the persistence of stereotypes and wrong beliefs (Snyder, 1981; Skrypnek and Snyder, 1982; Babcock et al., 2017). One mechanism for this persistence can be the existence of strong incentives for uniform behavior by which the stereotyped group cannot disprove the stereotype, leading to its persistence. Future work testing for the Failure to Anticipate Inference and to Plan in these settings would help understand the psychological underpinnings of these results.

Moreover, incorporating strategic concerns can help study the implications of our findings for information acquisition in collective decision-making. In this context, theoretical literature shows that under-investment in information can also arise from strategic incentives: in committees and electorates, rational agents free-ride on others' information acquisition (Persico, 2004; Martinelli, 2006), and in dynamic voting settings, control-sharing externalities reduce experimentation below the social optimum (Strulovici, 2010). Laboratory evidence from this setting is mixed: Bhattacharya et al. (2017) find that voters over-acquire information when signals are noisy, but under-acquire when signals are precise. Our results show that under-acquisition of information persists even when strategic motives are entirely absent: a single decision-maker, facing no free-riding incentive, still fails to screen. This suggests that insufficient information acquisition has both strategic and cognitive roots, and that failures of contingent thinking is a cognitive channel that may compound the evidence documented in collective settings.

As with any economic experiment, the introduction of a highly controlled environment comes at the cost of naturalism. While we believe our design captures the essence of the screening decision in an externally relevant way, future field work testing the role of contingent thinking in screening can further contribute to our understanding of this phenomenon. For example, while in naturally occurring scenarios screening is often intentional (e.g., hiring), the informational cost of underscreening can often be overlooked in situations in which preventing bad behavior takes a more salient role (e.g., prison policy). As another example, the external relevance of our results could be further tested by empirically quantifying the informational cost of underscreening in naturally occurring settings where the party making the inference differs from the one making the screening choice (e.g., as in the example above with prisons and inmates), including contexts where the policy choice is paternalistic. Future work could explore how this and other naturally occurring factors—that our experiment controls for—interact with our results.

Finally, the basic cognitive bias we explore can, in principle, relate to any setting in which

there is a risky choice that provides information. More work in this area is needed to figure out whether the cognitive phenomenon we study in this paper extends to other settings, like canonical search problems, stopping problems, and beyond. As an example of the breadth of settings that are related, Bernheim and Whinston, 1998 show that incomplete contracts may be optimal if they allow learning more about the counterparty than complete contracts. Future work could explore whether the Failure to Anticipate Inference and to Plan can lead people to ignore this channel and write too restrictive contracts.

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